



CHITKARA
UNIVERSITY

Industry Oriented Hands on Experience (22CS422)

SECURE INTERNSHIP & PLACEMENT MANAGEMENT

Vanshit

2210992509

Sehaj

2210992277

Satvik

2210992267

Supervised By: Dr. Gurpreet Singh

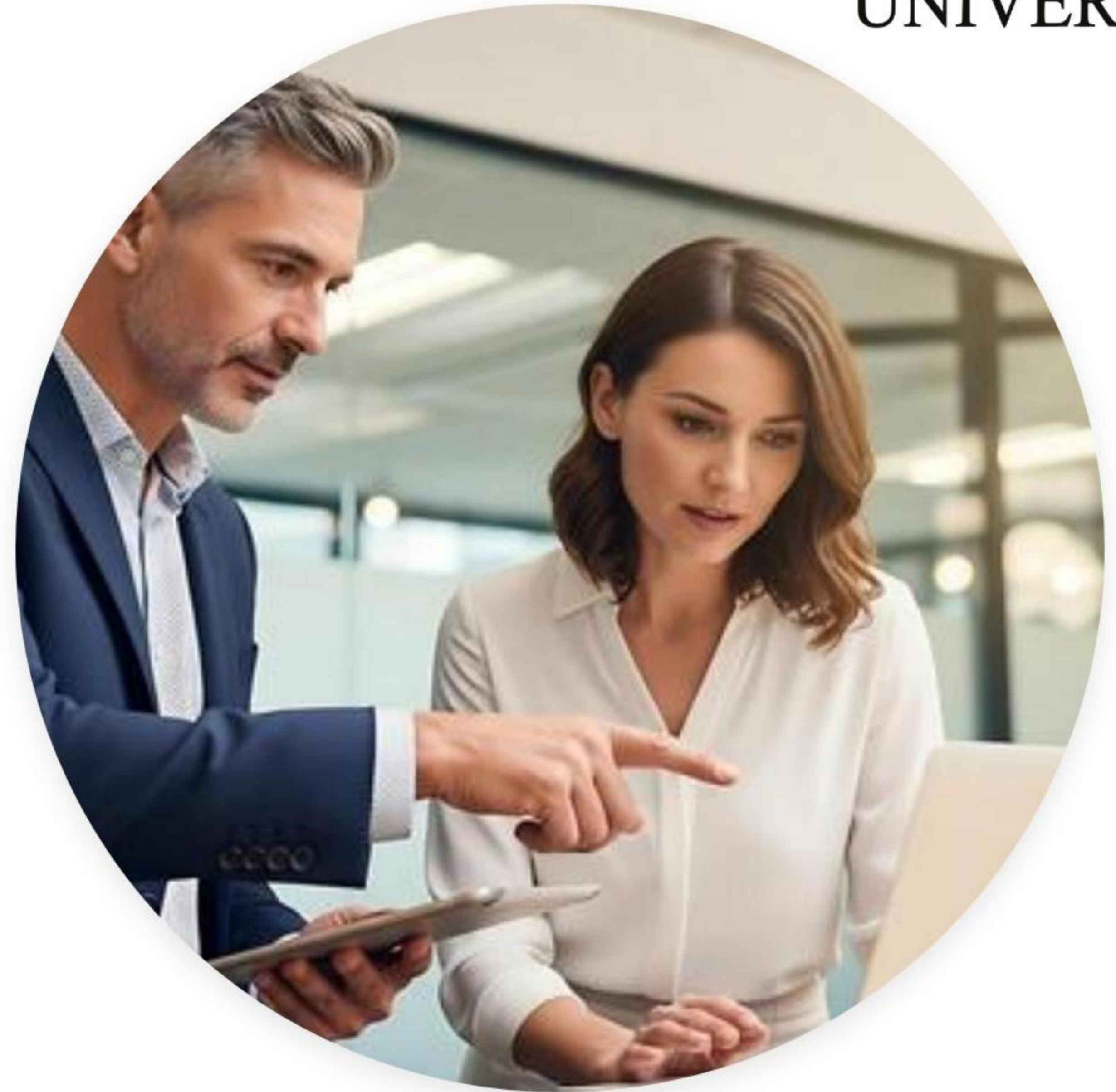
Department of Computer Science and Engineering, Chitkara University, Punjab

| Introduction

The Modern Solution

Campus recruitment is the pivotal bridge between academic excellence and industrial professional demand. Our system modernizes this transition.

- ✓ Integrated Full-Stack Ecosystem
- ✓ Real-time Application Tracking
- ✓ Data-Driven Recruitment Insights



Problem Statement



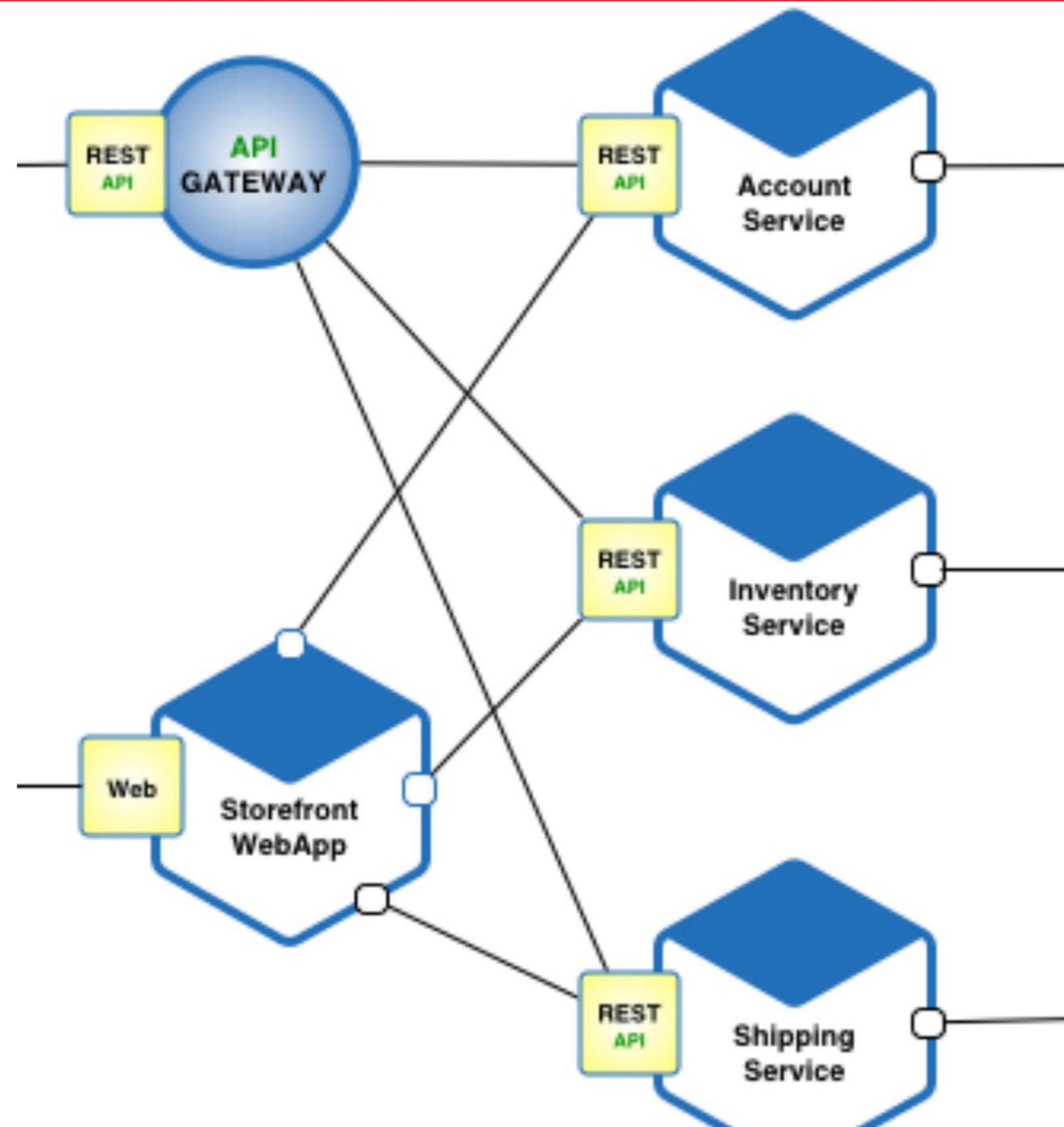
-  **Lack of Data Integrity:** Information scattered across fragmented spreadsheets and personal emails.
-  **Security Risks:** Sensitive student data and resumes exposed without proper encryption or access control.
-  **Inefficient Matching:** No automated way to score resumes against specific job descriptions efficiently.
-  **Transparency Gap:** Students lack real-time visibility into their application and shortlisting status.

| System Architecture

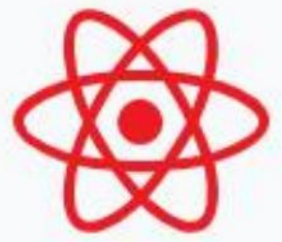
Microservice Design

Our platform is built using a modern, scalable microservice architecture ensuring high availability and seamless data flow between components.

The API Gateway manages client requests, routing them to specific services while maintaining centralized security and monitoring protocols.



Technology Stack



Frontend

React.js with Redux
Toolkit for high-
performance state
management and dynamic
UI.



Backend

Node.js & Express.js
providing robust REST
API services and secure
routing.



Database

MongoDB for scalable,
schema-less document
storage of resumes and
jobs.



Security

JWT authentication &
Bcrypt hashing for secure,
persistent user sessions.

Core Platform Features



Secure Auth

Role-based access control for Students and Recruiters with secure sessions.



AI Resume Scoring

Leveraging Google Gemini for instant score and strength analysis on resumes.



Admin Dashboard

Real-time applicant management and status updates for recruiters.

Platform Impact Metrics



75%

Faster Processing

Reduction in manual screening time.

100%

Secure Data

End-to-end encryption of identifiers.

Technical Deep Dive



Component	Technology	Primary Purpose
Frontend	React + Tailwind CSS	Modern, responsive User Interface
State Management	Redux Toolkit	Centralized data flow for Jobs/User
Backend API	Express + Multer	REST API & File processing
Storage	Cloudinary	Cloud hosting for resumes and images
AI Engine	Google Gemini API	Natural Language resume analysis



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FUNCTIONAL MODULES

Tailored experiences for every user segment

Student Workflow



-  **Smart Profile:** Dynamic resume building and skill tagging for students.
-  **Advanced Discovery:** Search internships by keyword, role, and location.
-  **One-Click Application:** Apply to multiple postings using saved credentials.
-  **Feedback Loop:** View real-time shortlisting status and AI score tips.

Recruiter Dashboard



-  **Company Setup:** Create and manage a branded profile for the organization.
-  **Job Posting:** Post roles with detailed requirements and salary bands.
-  **Applicant Tracking:** Scrutinize automated AI resume analysis.
-  **Status Management:** Bulk accept or reject candidates with notifications.

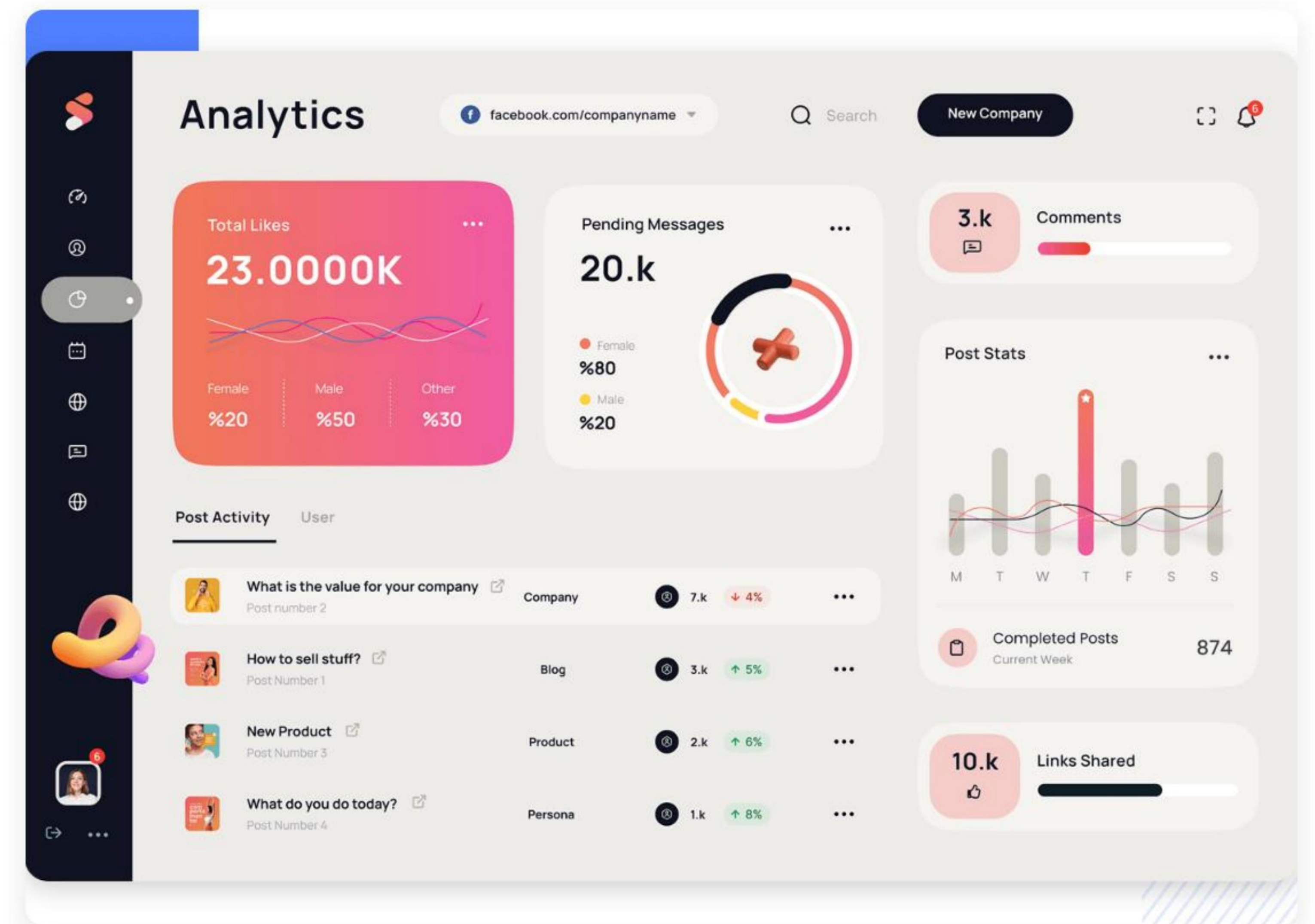
Interface Showcase

Modern UX Principles

Designed with a focus on accessibility and ease of use, ensuring students can focus on applications while recruiters manage pipelines efficiently.

 Fully Responsive Design

 Modern Dark/Light Themes

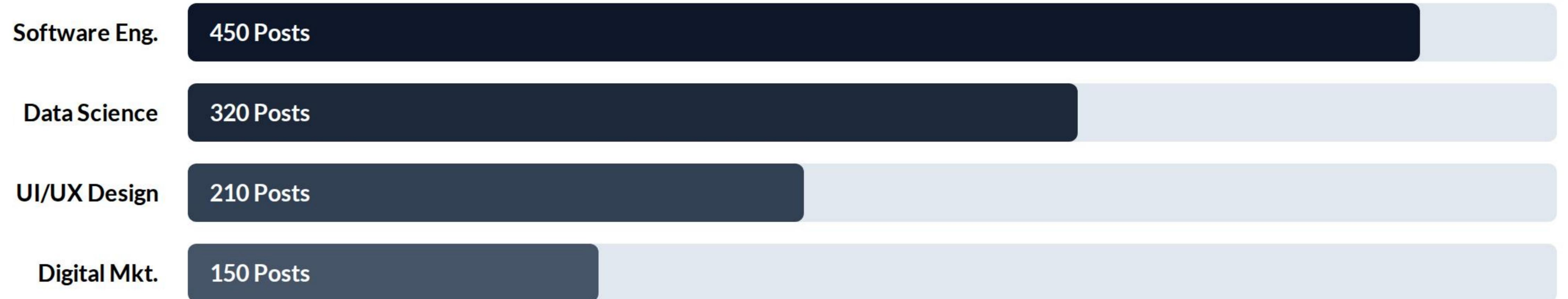


User Engagement Trends



Simulated growth in user applications post-AI integration.

Internship Role Distribution



Implementation Timeline



Security Architecture



JWT Tokenization

Stateless authentication using encrypted tokens stored in HTTP-Only cookies.



Bcrypt Hashing

Passwords never stored in plain text; salted and hashed with 10 rounds of security.



Private Routing

Custom middleware prevents URL manipulation and restricts unauthorized access.

CONCLUSION

Final takeaways and future outlook

The Road Ahead



Real-time Chat

Socket.io for recruiter
interaction.



Assessments

Built-in technical screening
tests.



Mobile Apps

Native iOS/Android job
notifications.



Forecasting

Predictive trends via
historical data.

THANK YOU

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